Import Data Using Transform Map

(Spreadsheet)

TEAM 12	NAME	NM ID
TEAM LEADER	S.MANIPRIYA	asaluau020224128022
TEAM MEMBER 1	G.MAREESWARI	asaluau020224128023
TEAM MEMBER 2	S.MARSUKKA SAFRIN	asaluau020224128024
TEAM MEMBER 3	S.VARSHINI	asaluau020224128036
TEAM MEMBER 4	M.AMMU	asaluau026624128001

DESIGN PHASE

1. System Design:

- The Design Phase defines the overall structure and working model of the Import Data using Transform Maps project in Service Now. The main purpose of the design is to provide a systematic method for importing employee information from an external spreadsheet into the Service Now platform.
- The system is designed to receive structured employee data in Excel format and process it through Import Sets and Transform Maps. The imported data is initially stored in an Import Set table, which acts as a temporary staging area. The Transform Map then connects the source fields with the corresponding fields in the target Employee Test table.

The overall design consists of the following major components:

- ❖ External Excel/Spreadsheet
- ❖ Import Set Table
- ❖ Transform Map
- ❖ Target Employee Test Table
- ❖ Field Mapping
- ❖ Coalesce Mechanism
- ❖ Reports
- ❖ Dashboard
- The design supports both new employee record insertion and updating of existing employee records. It also helps prevent duplicate records when the same employee data is imported repeatedly.

2. Architecture Design:

- The architecture of the proposed system follows a simple data-flow structure. Employee information is first prepared in a spreadsheet and then imported into Service Now.

Architecture Flow

External Spreadsheet → Import Set Table → Transform Map → Employee Test Table → Reports → Dashboard

2.1 External Spreadsheet:

- The external spreadsheet acts as the source of employee information. It contains structured data such flow,
- Employee ID

- Employee Name
- Email
- Department
- Location
- The spreadsheet is prepared with appropriate headers and sample employee records. It is then downloaded in Excel format and used as the input file for Service Now.

2.2 Import Set Table:

- The Import Set table is designed as a staging area for incoming employee data. The project creates an Import table named Employee Import.
- The Import Set receives the spreadsheet data before it is transferred to the final target table. This provides an intermediate layer where the imported data can be processed through the Transform Map.

2.3 Target Table:

- The target table is designed to store the final employee records. The project creates a custom table named Employee Test.
- The table contains the following fields

FIELD	Type
Employee ID	String
Employee Name	String
Email	String
Department	String
Location	String

- These fields provide the required structure for storing employee information in ServiceNow.

3. Transform Map Design:

- The Transform Map is the main processing component of the system. It defines how data from the Import Set table is transferred into the Employee Test table.The Transform Map is created with the following configuration:

Name : Sample Spreadsheet Import

Source Table : Employee Import

Target Table : Employee Test

For example:

Source Field	Target Field
Employee ID	Employee ID
Employee Name	Employee Name
Email	Email
Department	Department
Location	Location

- After completing the field mapping, the Transform Map is saved and executed using the Transform option.

4. Data Transformation Design:

- The data transformation process is designed to convert the imported spreadsheet information into usable Service Now records.

The transformation process consists of the following steps:

- Prepare employee data in Excel.
- Create the Employee Test target table.
- Create the Employee Import table.
- Load the spreadsheet into the Import Set table.
- Create the Transform Map.
- Map source fields with target fields.
- Execute the Transform operation.
- Verify the transformed records.
- After transformation, the employee records are created in the Employee Test table.
- The result is checked to ensure that the imported records are available correctly.

5. Coalesce Design:

- A major part of the system design is the Coalesce mechanism. Coalesce is used to identify existing records and avoid creating duplicate records.
- In this project, a field such as Employee ID can be configured as a Coalesce field in the Transform Map.
- When new data is imported, Service Now compares the Coalesce field with existing records.

The system follows this logic:

- ❖ If Employee ID exists → Update existing record
- ❖ If Employee ID does not exist → Insert new record
- This design is useful when employee information is imported multiple times. Existing employee records can be updated with new information instead of creating duplicate records.

For example, if an employee has the same Employee ID but a changed email address or name, the existing record is updated. If a completely new Employee ID is received, a new employee record is inserted.

- This makes the employee database more consistent and reliable.

6. Validation Design:

- Validation is included in the design to verify whether the transformation has been completed successfully.

After executing the Transform Map, the system checks:

- ❖ Import status
- ❖ Number of inserted records
- ❖ Number of updated records
- ❖ Number of ignored records
- ❖ Final records in the Employee Test table

For example, when four rows are imported, some records may be inserted while existing records may be updated. When the same data is imported again, the system can identify that the records already exist and avoid unnecessary duplicate insertion.

- The Transform History is used to verify the result of the transformation.

7. Report Design:

The system is designed with reports to provide a clear view of employee information.

Three major reports are created:

7.1 Employees by Department

This report displays the number of employees in each department.

Configuration:

- Table: Employee Test
- Type: Pie Chart
- Group By: Department
- Aggregation: Count

This report provides a visual representation of department-wise employee distribution.

7.2 Employees by Location

This report displays employee distribution based on location.

Configuration:

- Table: Employee Test
- Type: Bar Chart
- Group By: Location
- Aggregation: Count

The report helps users understand how employees are distributed across different locations.

7.3 Employee List Report

The Employee List Report provides detailed employee information in list format.

The report contains:

- Employee ID
- Employee Name
- Email
- Department
- Location

This report provides a complete view of the employee records stored in the Employee Test table.

8. Dashboard Design:

- The final part of the design is the creation of an Employee Analytics Dashboard.
- The dashboard provides a centralized view of the reports created from the Employee Test table.

The following reports are added to the dashboard:

1. Employees by Department

2. Employees by Location

3. Employee List Report

- The dashboard allows administrators and HR users to view employee information and related analytics from a single location.
- Reports can be added to the dashboard using the Add to Dashboards option. The system also supports adding widgets and configuring the dashboard according to the required presentation.

9. Overall Design Summary:

The complete system design provides an organized process for employee data migration and management in Service Now.

The designed workflow is:



- The design focuses on accurate data transfer, proper field mapping, duplicate prevention, record validation, and visual reporting.
- The use of Import Sets and Transform Maps provides a structured approach for importing external spreadsheet data into Service Now.
- The addition of Coalesce improves data consistency by preventing unnecessary duplicate records. Reports and dashboards provide centralized visibility into employee data.